# From Retrieval to Reasoning: How SOP Accessibility Shapes Interpretation in Regulated Work

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**Abstract**

Standard operating procedures (SOPs) in pharmaceutical manufacturing environments are legally binding documents — yet practitioners routinely fail to consult them, and prior research has focused on training gaps and motivational barriers rather than on the structural conditions of access. This paper argues that document accessibility is an upstream condition on interpretive practice: when retrieval is structurally difficult, practitioners do not merely search inefficiently — they act within a personally known subset of the procedural archive, constraining their behavioral repertoire before any formal task error occurs. I report a prototype-based pilot study conducted in a GMP quality control environment. I developed an external retrieval layer — a keyword-searchable metadata index over a corpus of 500 synthetic GMP SOPs — and evaluated it with $N = 20$ frequent SOP users using a think-aloud protocol. The prototype could not modify validated source files, a constraint constitutive of regulated manufacturing environments. Findings show that keyword-based access reduced exploratory retrieval time from approximately six minutes to approximately one minute, expanded the procedural range practitioners felt able to consult, and surfaced a category of invisible onboarding labor — informal SOP guidance provided by senior staff to resolve nomenclature ambiguities — that the quality system does not measure. I introduce SOP accessibility as a design problem specific to GMP-regulated environments, identify validated-archive constraints as a

novel design boundary for retrieval system research, and offer a theoretical framing that connects HCI concerns about document navigation to broader questions about how responsibility is structured in regulated sociotechnical organizations.

**Keywords:** SOP accessibility, document retrieval, pharmaceutical manufacturing, GMP, interpretive variability, invisible work, regulated environments

# 1 Introduction

Standard operating procedures (SOPs) are among the most consequential documents in regulated manufacturing. In pharmaceutical production environments governed by Good Manufacturing Practice (GMP) regulations, SOPs define how every process must be performed, and deviation from them can constitute a regulatory violation. Yet the practical challenge of consulting an SOP at the moment it is needed has received almost no attention in the HCI literature. Prior work on SOP non-compliance has focused on training gaps, motivational barriers, and procedure design — not on the structural conditions under which practitioners access documents in the first place (Krockenberger et al., 2008; Hales & Pronovost, 2006).

This matters because access conditions shape interpretation. When practitioners cannot locate a relevant procedure quickly — because archives are stored as scanned PDFs that block keyword search, because titles are too similar to distinguish, or because time pressure makes trial-and-error retrieval costly — they do not simply fail to comply. They navigate under constraint: working from memory, deferring judgment, or asking colleagues. This substitutive work introduces interpretive variability before any formal task error occurs, and before responsibility is later assigned. It is, in the terms of Star & Strauss (1999), invisible — unmeasured by quality systems and unattributed to the structural conditions that produced it.

I argue that document accessibility is an upstream condition on interpretation in regulated work environments: retrieval burden shapes the range of procedures a practitioner encounters, compares, and can act on with confidence. This argument extends Suchman's

(1987) account of situated action to the domain of procedural document use, and identifies a design opportunity that prior HCI work has not addressed — not because it is unimportant, but because the GMP archive presents a design constraint with no precedent in existing retrieval system research: validated source files cannot be modified, meaning that any retrieval improvement must operate externally, over metadata, without altering the documents themselves.

This paper reports a prototype-based pilot study in a pharmaceutical GMP environment. I developed an external retrieval layer — a keyword-searchable metadata index — over a corpus of approximately 500 procedural documents, and evaluated it with $N = 20$ frequent SOP users using a think-aloud protocol. The prototype served simultaneously as a design intervention and as an analytic probe: by observing how participants navigated the system, I was able to make upstream retrieval burden visible and tractable in a way that conventional error analysis cannot. Findings show that reducing retrieval effort shifted participant attention from document navigation to document interpretation — and that the friction revealed was structural rather than individual, traceable to archive design rather than user failure.

The contributions of this work are threefold. First, I introduce SOP accessibility as a design problem within GMP-regulated environments, identifying validated-archive constraints as a novel design boundary for retrieval system research. Second, I demonstrate that a prototype retrieval layer can reduce navigational burden without modifying validated source files, preserving document integrity under regulatory requirements. Third, I offer a theoretical framing — accessibility as an upstream condition on interpretive variability — that connects HCI concerns about document navigation to broader questions about how responsibility is structured in regulated sociotechnical organizations.

The remainder of the paper is structured as follows. Section 2 reviews related work on procedural usability, document accessibility, and accountability in regulated systems. Section 3 describes the study design and method. Section 4 presents findings. Section 5 discusses implications for compliance-constrained design. Section 6 concludes.

# 2 Related Work

## SOP Non-Use and the Limits of Procedural Infrastructure

Prior research has documented that practitioners routinely fail to consult standard operating procedures at the point of action. Krockenberger et al. (2008) found that 18.4% of clinical trial coordination staff sought a colleague's guidance rather than retrieving the relevant SOP, and concluded that the primary barrier was inadequate retrieval system design. This diagnosis — that better search infrastructure would resolve SOP non-use — has shaped subsequent system development in regulated environments. A broader literature on procedural adherence confirms that non-compliance with written procedures is rarely a matter of individual negligence; rather, it reflects structural conditions including inaccessible documentation, outdated archives, and the gap between procedures as designed and tasks as performed (Hales & Pronovost, 2006). Yet both lines of work leave a critical question unaddressed: even when retrieval is improved, does better access translate into more consistent interpretation? The present study takes up precisely this gap, examining not whether practitioners can locate SOPs, but how variability in access conditions shapes the interpretive work that follows.

## Professional Document Work and the Persistence of Retrieval Difficulty

The difficulty of locating authoritative documents is not unique to pharmaceutical manufacturing. Sellen & Harper (2002) demonstrate that digital transitions do not resolve the retrieval problems of document-intensive work; rather, they shift the role of documents without eliminating the navigational demands that practitioners face. Their analysis of document affordances shows that how documents are stored and accessed fundamentally shapes how they are used — a finding that applies with particular force in GMP archives, where scanned procedural files block keyword search and force reliance on folder memory. Harper's (1998) ethnographic study of documentation practices at the IMF similarly shows that documents do not operate as neutral carriers of information: their accessibility

structures the work that surrounds them, and gaps in retrieval support create invisible friction that is rarely captured in organizational accounts. Russell-Rose et al. (2018) extend this argument across healthcare, legal, and patent work contexts, demonstrating that retrieval difficulty is a structural feature of professional document environments, not a correctable surface problem. The present study examines how this structural difficulty manifests in GMP contexts, where the constraint of validated archives uniquely restricts what retrieval interventions are permissible.

## Situated Action and the Inaccessible Procedure

The theoretical basis for treating access conditions as consequential for interpretation comes from Suchman's (1987) account of situated action. Suchman argues that human action is not plan-driven but shaped by the immediate conditions of its execution — what is visible, reachable, and actionable in the moment of practice. Applied to SOP use, this framework implies that a procedure's formal existence does not determine its practical effect; what matters is what is accessible to the practitioner at the moment action is required. Unstable or inconsistent access to SOPs thus becomes not a logistical inconvenience but a structural source of interpretive variability — a point prior HCI work on procedural documents has noted but not operationalized in the context of regulated manufacturing.

## Invisible Work and Upstream Labor

When SOPs are inaccessible, practitioners do not simply fail to act; they perform substitutive work — asking colleagues, reconstructing procedures from memory, or deferring action. Star & Strauss (1999) show that work of this kind tends to become invisible: it is not captured by formal indicators, not measured by quality systems, and not attributed to the access failure that produced it. In regulated manufacturing, this invisibility has direct implications: the retrieval burden created by inaccessible documentation sits upstream of any formal error record, and is therefore structurally absent from CAPA analysis and audit review. The prototype-based study reported here makes this upstream burden

legible — treating it not as individual user difficulty but as a structural condition of the archive that retrieval interventions can address without modifying validated source files.

## Think-Aloud Methods and Prototype-Based Evaluation

The methodological approach in this study draws on an established body of HCI research on think-aloud evaluation and prototype-based inquiry. Ericsson & Simon (1993) provide the theoretical foundation for think-aloud protocols as a data collection method: concurrent verbalization surfaces the cognitive processes underlying task performance without substantially altering their course, making it a particularly appropriate method for studying navigation and search behavior under realistic task conditions. Nielsen (1994) operationalizes this approach within usability engineering, demonstrating that think-aloud studies with relatively small participant samples reliably surface the majority of usability problems in an interface. Jaspers et al. (2004) apply this method in a regulated clinical context directly analogous to the present study: using think-aloud protocols and video recording to analyze how clinicians searched through paper-based patient records, they developed a prototype interface that reduced retrieval burden without altering source documents — a design constraint that mirrors the validated-file requirement in pharmaceutical GMP environments. Hutchinson et al. (2003) further establish the logic of using a prototype as both an intervention and an analytic probe: a technology probe is simultaneously a design artifact and a research instrument, making it possible to study how access conditions shape behavior rather than simply to evaluate whether a given interface is usable. The present study applies this dual function to the GMP context, using a prototype retrieval layer to make visible the upstream conditions — structural inaccessibility, interpretive friction, substitutive labor — that conventional usability studies would not surface.

Prior work has not examined SOP accessibility as a design problem within pharmaceutical GMP environments, where document integrity constraints uniquely restrict retrieval system design. This study addresses that gap directly.

# 3 Method

## Study Context

This study was conducted in the quality control department of a pharmaceutical manufacturing facility operating under 21 CFR Part 210/211 Good Manufacturing Practice regulations. The procedural archive used in this study consisted of 500 synthetic quality control SOPs generated programmatically in Python. Because actual facility SOPs are subject to non-disclosure obligations, the corpus was constructed to reflect the structural and terminological characteristics of real GMP quality control documentation — including document identifiers, process category conventions, procedural language, and keyword distributions typical of QC operations — without reproducing proprietary content. Documents were rendered as PDFs and parsed to extract metadata for indexing. This approach preserves the ecological validity of the retrieval problem — practitioners searched through a corpus with the naming conventions, terminological overlap, and navigational ambiguity characteristic of real GMP archives — while maintaining confidentiality requirements. The validated-file constraint that defines the core design boundary of this work applies equally to synthetic and live corpora in GMP environments.

## Phase 1: Metadata Mapping and Keyword Analysis

I conducted a systematic analysis of the full procedural archive to construct a searchable metadata index. For each document, I extracted the document identifier, title, process category, and keywords derived from the document content. Where keyword coverage was sparse — common in older scanned documents with non-standardized titling conventions — I supplemented extraction with manual annotation based on process area and task type. I also mapped structural overlap across the corpus: documents sharing similar titles, overlapping keyword sets, or belonging to the same process category were flagged as high-ambiguity clusters. This analysis identified the specific points in the archive where practitioners faced the greatest navigational difficulty — not from individual document complexity, but from the structural similarity of adjacent procedures.

## Phase 2: Prototype Development

Based on the metadata index, I developed an external retrieval layer as a Streamlit web application. The prototype allows practitioners to enter task-relevant keywords — terms describing what they are trying to do, rather than document titles or identifiers — and returns a ranked list of relevant SOPs with document identifiers, process categories, and contextual snippets. The system operates entirely over the metadata index; it does not modify, version, or interact with the validated source files in any way. Human interpretive authority is preserved: the system surfaces candidate documents, but practitioners retain full responsibility for reading, applying, and judging the procedures returned.

The prototype was designed around three principles derived from the accessibility analysis: (1) keyword-based access that decouples retrieval from archive naming conventions; (2) result cues (document IDs, category labels, snippets) that reduce early-stage uncertainty without resolving interpretation; and (3) a no-modification constraint that keeps the system within GMP compliance boundaries.

## Phase 3: Pilot Evaluation

I evaluated the prototype with $N = 20$ participants recruited from the same GMP quality control team. All participants were frequent SOP users with daily procedural document search experience. Participants were asked to complete a series of realistic search tasks using the prototype — tasks drawn from their own reported search scenarios — while thinking aloud throughout. Sessions were conducted individually; a researcher observed and took field notes without intervening in participant navigation. Approximate retrieval times were recorded informally as a descriptive measure of navigational effort; these are exploratory observations rather than controlled measurements.

Following the search tasks, participants completed a brief post-task questionnaire assessing ease of finding relevant sections, usefulness of result cues, overall intuitiveness, and willingness to reuse the system in similar work contexts. Open-ended prompts invited participants to describe remaining friction points and to articulate how the prototype differed from their current search practice.

Think-aloud data were analyzed following Ericsson & Simon's (1993) protocol analysis approach, with particular attention to moments of hesitation, uncertainty, and navigational substitution — instances where participants paused, reworded queries, or expressed doubt about whether a returned result was the correct procedure.

### Scope and Limitations

This study is exploratory and pilot in scope. Participants were recruited from a single team within a single facility, which may have inflated usability and acceptance ratings relative to a more diverse sample. Retrieval time observations are informal and should not be interpreted as controlled performance measurements. Results indicate usability promise and identify structural friction patterns, but do not establish long-term adoption effects, productivity outcomes, or downstream decision quality.

## 4 Findings

### Retrieval Burden as a Felt Cost

Participants consistently described SOP retrieval as a meaningful burden in their daily work — not a minor inconvenience but a recurring cost that competed with task execution. Several noted that locating the correct procedure under time pressure was itself effortful work: one participant described the experience as "dying of busyness while SOP search is also work." This framing positions retrieval not as a precondition for work but as a component of the work itself — labor that is expended before any procedural judgment begins.

The prototype reduced this burden in observed use. Exploratory retrieval times dropped from approximately six minutes under current conditions to approximately one minute with the prototype, though these are informal observations rather than controlled measurements. Participants reported that keyword-based access allowed them to enter task terms rather than document titles or identifiers, reducing reliance on archive naming familiarity. Result cues — document identifiers, category labels, and contextual snippets —

were rated as useful (4.5/5) in helping narrow the search space before opening documents. Overall intuitiveness was positive, and all participants indicated willingness to reuse the system in similar task contexts.

## Accessibility Constraints on Behavioral Repertoire

The most consequential finding concerned the scope of action that practitioners allow themselves. Participants described a pattern in which the practical difficulty of locating unfamiliar SOPs leads them to operate within a personally known subset of the archive — consulting procedures they have used before rather than searching for procedures that might be more directly applicable. In the words of one participant: when you cannot find a SOP easily, you end up acting within the SOPs you already know.

In GMP environments, this has direct regulatory implications. SOPs function as the formal basis for action: they define what practitioners are authorized to do and provide the documented justification for any action taken. When accessibility constraints limit the set of procedures a practitioner encounters, they also limit the range of authorized actions available to them — not because the procedures do not exist, but because they are not reachable. The prototype made a wider procedural landscape accessible, and participants reported that this expanded their sense of what actions were available and supportable.

## Nomenclature Ambiguity and Onboarding Labor

A second finding concerned the distribution of retrieval burden across experience levels. Senior practitioners described a recurring onboarding burden: new staff, unfamiliar with equipment naming conventions and archive organization, routinely struggle to distinguish between similarly named procedures. A representative example: a senior practitioner directing a new hire to consult the UPLC SOP found that the new hire searched for the HPLC SOP instead, not recognizing the distinction between the two instrument types. Under current archive conditions, resolving this kind of ambiguity requires intervention from a more experienced colleague — creating a pattern of informal knowledge transfer that is not captured in any formal quality metric.

The prototype partially addressed this. Keyword-based retrieval reduced dependence on exact nomenclature, and result snippets provided contextual differentiation to help less experienced users distinguish between adjacent procedures. Participants anticipated that this could reduce the informal SOP-guidance labor that senior practitioners currently provide — labor that, consistent with Star & Strauss (1999), is invisible to the quality system and unmeasured by any existing indicator.

### Remaining Friction

Not all friction was resolved. Participants noted that no-result states remained ambiguous: when a search returned nothing, it was unclear whether the issue was a wording mismatch or a genuine gap in archive coverage. This points to a metadata design problem rather than an interface problem — the current index does not distinguish between "this procedure exists under a different keyword" and "this procedure does not exist."

## 5 Discussion

### Accessibility as an Upstream Condition on Interpretive Practice

The central finding of this study is not that a retrieval tool reduces search time — though it does — but that access conditions shape the behavioral repertoire practitioners allow themselves in regulated work. When SOP retrieval is structurally difficult, practitioners do not simply search less efficiently; they act within a personally known subset of the procedural archive, consulting familiar procedures rather than searching for those that might be more applicable. In GMP environments, where SOPs are the formal basis for authorized action, retrieval constraints translate directly into constraints on what a practitioner can justify doing. The interpretive burden begins not at the moment of reading a procedure but upstream, at the moment of deciding which procedures are reachable.

This finding extends Suchman's (1987) account of situated action in a direction that prior HCI work has not pursued. Suchman established that action is shaped by immediate

conditions of execution rather than by pre-specified plans. Applied to SOP use, this implies that the formal existence of a procedure does not guarantee its practical effect — what matters is what is accessible at the moment action is required. Our findings show the consequences of this when accessibility is structurally constrained: practitioners develop implicit repertoires bounded by what the archive makes findable, and these repertoires persist even when more applicable procedures exist elsewhere in the corpus.

## Invisible Work Across Experience Levels

The onboarding finding illustrates the mechanism that Star & Strauss (1999) describe as invisible work. Senior practitioners' informal SOP guidance labor is real, recurring, and consequential: it consumes experienced practitioners' time, creates informal dependencies between staff, and shapes how new practitioners develop their procedural knowledge. It is invisible in the specific sense that Star and Strauss identify: it does not appear in any quality metric, is not captured by deviation reports or CAPA records, and is not attributed to the archive structure that produces it.

The prototype partially surfaced this invisibility by making nomenclature disambiguation possible without colleague intervention. This moves a category of informal coordination labor into the design layer, where it can be addressed structurally rather than managed interpersonally. The broader implication is that retrieval system design in regulated environments is not only a usability problem but an organizational one — a question of how labor is distributed, made visible, and attributed across a workforce.

## Design Implications for Regulated Document Environments

This study identifies a design space with no direct precedent in existing retrieval system research: external retrieval layers that operate over validated archives without modifying source files. The validated-file constraint is constitutive of GMP environments: any design intervention that requires modifying, tagging, versioning, or restructuring validated documents is, in a regulatory sense, a change to the documents themselves — and therefore subject to full change control. This means that the design of retrieval support

in pharmaceutical manufacturing is fundamentally different from retrieval design in other professional document environments, and that lessons from healthcare IT, legal search, or enterprise knowledge management transfer only partially.

The external metadata layer approach demonstrated here preserves document integrity while shifting the access problem from the archive to the retrieval interface. Several design principles emerge from the pilot evaluation. First, keyword-based access that decouples retrieval from archive naming conventions reduces the expert knowledge required to search effectively. Second, result cues that provide enough context to distinguish between adjacent procedures without requiring practitioners to open each document reduce the cost of navigating high-ambiguity clusters. Third, no-result transparency — making it clear whether a null result reflects a wording mismatch or a genuine archive gap — remains an unresolved design challenge that future iterations should address.

## Limitations and Future Work

Several limitations bound the interpretation of these findings. The synthetic corpus has less terminological and structural variation than a live archive would, which may have both underrepresented navigational difficulty and reduced result discriminability. Participant recruitment from a single team within a single facility introduces potential for inflated usability and acceptance ratings. The study lacks a comparative baseline, as no prior HCI tooling existed in this context. Think-aloud sessions were exploratory and session-based; they do not support claims about long-term adoption, sustained behavior change, or downstream effects on interpretation quality or regulatory compliance.

The most important direction for future work is deployment with a live procedural archive. A synthetic corpus can reveal structural friction patterns and validate the retrieval approach in principle, but the full implications — including effects on behavioral repertoire, onboarding efficiency, and the distribution of invisible labor — cannot be assessed without real-world use. Future work should also examine the interpretive consequences of improved access: if practitioners reach a wider range of procedures, does this lead to more consistent interpretation, or does it introduce new forms of ambiguity by expanding the comparison

set? The no-result problem identified here also points toward a design question that retrieval system research has addressed in other contexts but not in validated-archive environments: how to make absence legible without modifying the documents whose absence is being signaled.

# 6 Conclusion

This paper introduced SOP accessibility as an upstream condition on interpretive practice in regulated manufacturing environments. When retrieval is structurally difficult, practitioners act within a personally known subset of the procedural archive — not because they lack compliance motivation, but because the archive does not make a broader range of procedures reachable. This constraint shapes behavioral repertoire before any formal task error occurs, and before responsibility is later assigned.

The prototype-based pilot study demonstrated that an external retrieval layer — operating over metadata without modifying validated source files — can reduce this upstream burden: expanding the procedural range practitioners feel able to consult, surfacing invisible onboarding labor, and shifting attention from document navigation to document interpretation. These findings position SOP accessibility as a design problem distinct from procedural readability, and identify the validated-archive constraint as a novel design boundary specific to GMP-regulated environments.

More broadly, this work connects HCI concerns about document navigation to organizational questions about how responsibility is structured in regulated sociotechnical systems. Retrieval constraints are not merely usability problems; they shape the interpretive conditions under which accountability is later assigned, making document accessibility a structural precondition of equitable and auditable regulatory compliance.

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